

SUBHAM KARMAKAR

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Education

Adamas University

B.Tech in Computer Science and Engineering | Kolkata, West Bengal, India

Aug 2024 – May 2028 (Expected)

CGPA: 9.17 / 10

Technical Skills

Languages: JavaScript, TypeScript, Python, C, C++, SQL

Frontend: React.js, Next.js, Tailwind CSS, GSAP, Framer Motion, Redux Toolkit, HTML5/CSS3

Backend: Node.js, Express.js, FastAPI, REST API, WebSockets, Socket.io, Stripe

Databases: PostgreSQL, pgvector, MongoDB, Supabase, Neon, Redis

Testing: Jest, Vitest, pytest

Tools & DevOps: Git, GitHub, Docker, Docker Compose, GitHub Actions, Nginx, PM2, Vercel, Render, Railway, Sentry, Linux, CI/CD

Work Experience

Adamas University

Nov 2025 – Present

Moodle LMS Cloud Maintainer (Part-time)

Kolkata, India

- **Maintained 99%+ platform uptime** for **1,000+** students and faculty by administering the cloud-hosted Moodle LMS, managing incident response workflows, and implementing proactive system health monitoring.
- **Reduced** faculty support tickets by **~40%** by redesigning role-permission flows, documenting configuration standards, and eliminating the most common misconfiguration errors across the course onboarding process.
- **Streamlined** course setup for **100+** faculty members by standardizing plugin configuration and onboarding workflows, cutting average per-course setup time by approximately half.

Ryze AI

Jan 2026 – Apr 2026

Frontend Developer Intern

Remote

- **Accelerated** feature release velocity across multiple sprint cycles by shipping production UI components in React.js, TypeScript, and Tailwind CSS — contributing directly to client-facing feature launches within tight delivery windows.
- **Reduced** average data-fetch latency by **~30%** by integrating REST API endpoints with optimized request patterns and component-level caching strategies, in close collaboration with backend engineers.
- **Maintained zero missed sprint commitments** across the full engagement by actively participating in standups, code reviews, and retrospectives while sustaining a consistent PR merge velocity throughout.

Projects

CyCoder | Next.js, TypeScript, PostgreSQL, Tailwind CSS, Docker, Nginx, PM2 | cycoders.in

2025

- **Scaled** Adamas University's developer community to **100+ active users** with week-over-week retention by engineering a gamified XP-progression system, leaderboards, and multi-role auth (Admin/Member) — turning passive signups into daily-active contributors powering Signifiya.
- **Eliminated 100%** of manual attendance overhead for **100+** members by building a QR-code check-in system with real-time CRUD controls, admin dashboards, and exportable logs.
- **Achieved zero-downtime deployments** by containerizing the full TypeScript stack with Docker Compose, self-hosting on a VPS behind Nginx, PM2 process management, and GitHub Actions CI/CD.

Noted | Next.js, TypeScript, Node.js, WebSockets, PostgreSQL, Redis, Stripe, LLM APIs | notezero.vercel.app

2025

- **Reduced** manual note organization effort by **~95%** by building a fully agentic pipeline where LLM calls autonomously classify intent, enrich content, generate structured tags, and surface context-aware linked notes as the user writes.
- **Achieved zero data loss** under concurrent multi-user edits by implementing real-time WebSocket collaboration with Redis-backed pub/sub messaging and an operational-transform-inspired conflict resolution strategy.
- **Cut** context-switching from minutes to seconds with Redis-accelerated AI retrieval surfacing related notes inline; Stripe handles subscription billing for premium access tiers.

CrackCoded | Next.js, TypeScript, FastAPI, Python, PostgreSQL, pgvector, Render, LLM APIs | crackcoded.vercel.app

2026

- **Delivered 340% faster learning acceleration** vs. unguided practice by building a Python/FastAPI backend — deployed on **Render** — that queries **pgvector** embeddings on PostgreSQL to surface SM-2-scheduled problem queues from LeetCode, Codeforces, and HackerRank; Jest test suite covers critical recommendation logic.
- **Reduced** average time-per-problem by **67%** by integrating an AI hint engine that analyzes the user's approach, pinpoints where reasoning breaks down, and delivers skill-calibrated personalized guidance — replacing generic editorial walkthroughs.
- **Achieved 72% average topic completion rate** across users by eliminating random practice and replacing it with weakness-targeted pgvector similarity queues that close concrete DSA gaps.

Achievements & Certifications

Claude Code 101 — Anthropic Education, Jun 2026 | verify.skilljar.com/c/y7qppxfzwffs

Introduction to Model Context Protocol — Anthropic Education, Jun 2026 | verify.skilljar.com/c/s4t3vecwu84e

Smart India Hackathon 2025 — Internal Finalist, Top **10** in Software Category out of **150** competing teams

Certified Full Stack Web Developer — Udemy, **97** hours (ID: UC-f427f3a7-351a-41fc-91e4-510256962222)